Meta-analytic structural equation modeling as a framework for estimating between- and within-individual effects in multilevel dynamic models

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Abstract

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Keywords: vector autoregressive model, multilevel model, random-effects meta-analysis, mixed-effects meta-analysis, distal outcomes

Empirical Example

To illustrate the utility of the MetaVAR framework, we applied it to intensive longitudinal data from adults diagnosed with generalized anxiety disorder (GAD) or major depressive disorder (MDD), collected by Fisher et al. (2017).

Participants completed approximately 30 days of ecological momentary assessment (EMA), responding to smartphone prompts multiple times per day about their momentary emotional states, avoidance behaviors, and cognitive symptoms. This dataset provides an ideal context for demonstrating MetaVAR because it includes (a) rich, high-frequency within-person time series of affective processes and (b) a detailed set of between-person clinical variables assessed.

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This research was made possible by the Prevention and Methodology Training Program (PAMT) funded by a T32 training grant (T32 DA017629 MPIs: J. Maggs & S. Lanza) from the National Institute on Drug Abuse (NIDA).

Your grants.

Computations for this research were performed on the Pennsylvania State University's Institute for Computational and Data Sciences' Roar supercomputer.

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Affective Processes in MDD and GAD

The persistence and temporal organization of emotional states—rather than simply the levels of these states—play a central role in the maintenance of internalizing psychopathology. A growing literature links negative affect inertia to depression, anxiety, and impaired emotion regulation (Houben et al., 2015; Kuppens, Allen, & Sheeber, 2010). Conversely, positive affect persistence supports recovery from stress and predicts psychological well-being (Hollenstein, 2015). Cross-lagged relations between positive and negative affect, particularly PA $\rightarrow$ NA dampening and NA $\rightarrow$ PA suppression, are theorized to index resilience or vulnerability in emotion-regulation systems (Bringmann et al., 2016; Kuppens, Oravecz, & Tuerlinckx, 2010).

EMA-Derived Within-Person Variables

From the EMA items, we created two core within-person processes: (a) Negative Affect (NA): "felt down or depressed", "felt hopeless", "loss of interest or pleasure", "felt worthless or guilty", "felt worried", "felt irritable"; and (b) Positive Affect (PA): "felt positive", "felt content", "felt enthusiastic", "felt energetic". These groupings reflect Fisher et al.'s (2017) conceptual clusters of distress and positive emotional functioning. Composites were computed at each EMA prompt to obtain dynamic time series suitable for idiographic modeling.

We estimated a two-variable VAR(1) for each participant, yielding person-specific dynamic parameters indexing (a) inertia (AR(NA), AR(PA)) and (b) positive–negative affect coupling ($\mathrm{PA}_{t-1} \rightarrow \mathrm{NA}_t$, $\mathrm{NA}_{t-1} \rightarrow \mathrm{PA}_t$). We also estimated person-specific measurement intercepts, which capture each individual's typical (set-point) level of NA and PA (NA(Intercept), PA(Intercept)). Together, these idiographic parameters allow us to test clinically relevant mechanisms linking affective set-points and regulatory dynamics to downstream functioning.

Baseline Covariate

We used baseline primary diagnosis (MDD vs. GAD) as between-person predictor of individual differences in affect dynamics. Diagnosis was selected because MDD and GAD are theorized to differ in core affective dysfunction: MDD is characterized by diminished positive affect and anhedonia (i.e., a lower hedonic set point), whereas GAD is characterized by pervasive worry that sustains negative affect and dampens shifts in emotional state (TODO: CITATION HERE). Accordingly, diagnosis was modeled as a predictor of (a) person-specific measurement intercepts (NA(Intercept), PA(Intercept)) and (b) person-specific VAR(1) parameters indexing inertia (AR(NA), AR(PA)) and positive–negative affect coupling ($\mathrm{PA}_{t-1} \rightarrow \mathrm{NA}_t$, $\mathrm{NA}_{t-1} \rightarrow \mathrm{PA}_t$).

Distal Outcomes

To capture short-term clinical status after the period in which VAR dynamics were estimated, we averaged NA, PA, and AV items from the final several days of EMA. These distal outcomes allow us to examine whether person-specific dynamic parameters predict:

higher subsequent negative affect; lower subsequent positive affect; or persistent or escalating avoidance. This operationalization maintains all constructs within a unified EMA system and mirrors clinical models of short-term symptom change and daily functioning.

Research Questions and Hypotheses

Using this combined baseline–dynamics–distal structure, we examine three sets of questions:

RQ1: What are the typical patterns and individual differences in dynamic emotional and behavioral processes among adults with GAD and MDD?

On average, we expect moderate positive inertia for NA and PA, and significant coupling between affect and avoidance behaviors. However, substantial between-person variability should emerge in all parameters, consistent with prior idiographic network findings.

RQ2: Do baseline clinical characteristics predict individual differences in dynamic parameters?

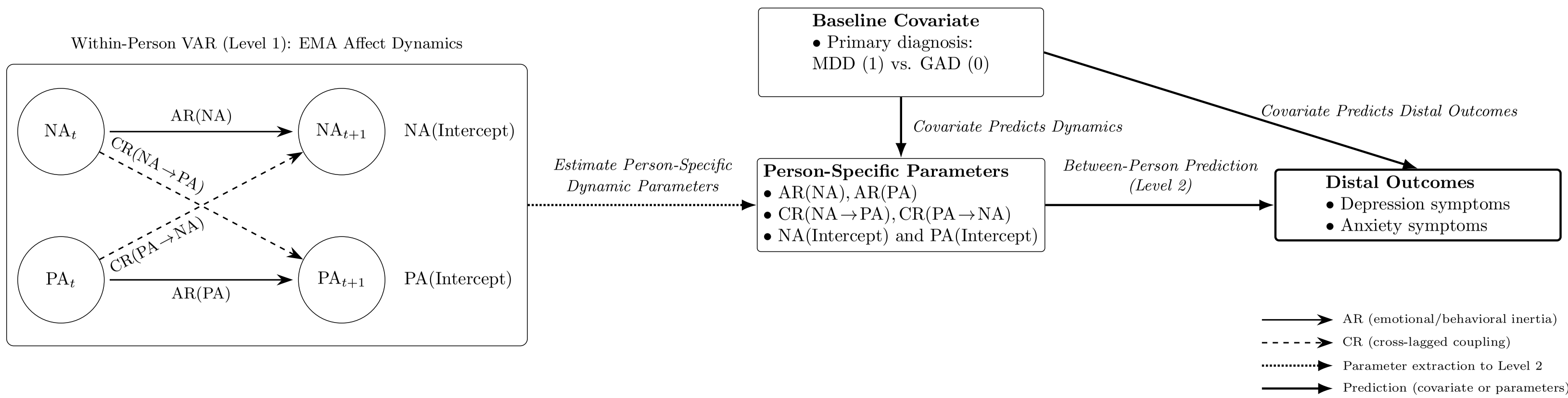
Individuals' baseline clinical characteristics are hypothesized to systematically predict variation in their idiographic dynamic parameters. Specifically, we expect that participants with higher clinician-rated depressive and anxious symptom severity, as indexed by HAM-D and HAM-A scores, will exhibit more maladaptive dynamic profiles. These profiles include stronger negative affect inertia, weaker recovery of positive affect, and tighter feedback loops between affect and avoidance. We further anticipate that diagnostic distinctions will be reflected in dynamic differences: individuals with primary MDD, compared with those with primary GAD, are expected to show reduced persistence of positive affect and stronger suppression of positive affect following increases in negative affect. In addition, we expect that the presence of comorbid social anxiety disorder will be associated with more pronounced avoidance-related dynamics, including stronger avoidance→negative affect cross-lags and greater avoidance inertia. Together, these predictions align with contemporary dimensional models of internalizing psychopathology, which emphasize the centrality of avoidance, diminished positive affect, and impaired emotion-recovery processes in maintaining distress.

RQ3: Do dynamic parameters predict later EMA-based outcomes, over and above baseline severity?

We next examine whether person-specific dynamic parameters prospectively predict later emotional and behavioral functioning, above and beyond baseline clinical severity and early EMA-derived covariates. We hypothesize that individuals exhibiting stronger negative mood inertia, greater avoidance inertia, and more pronounced avoidance→negative mood coupling will show worse distal outcomes, reflected in higher levels of negative mood and avoidance during the final days of EMA. In contrast, individuals with more adaptive dynamic profiles—such as stronger positive mood persistence and stronger positive → negative mood dampening—are expected to report more favorable distal functioning, including

Figure 1

MetaVAR Framework Linking Within-Person Affect Dynamics to Between-Person Covariates and Distal Outcomes



Note: AR = autoregression; CR = crossregression; NA = negative affect; PA = positive affect; MDD = major depressive disorder; GAD = generalized anxiety disorder.

higher late positive mood and lower late negative mood. This set of hypotheses evaluates whether idiographic emotion–behavior dynamics serve as mechanisms connecting baseline symptom burden to short-term clinical trajectories within a fully ecological, EMA-based framework.

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